

Internet of Things and Vehicular Networks (TP-546)

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Platoon Networks (Platooning)



A platoon is a group of vehicles that can travel very close together safely at high speed. The vehicles are guided by the first car.

Each vehicle communicates with the other vehicles in the platoon. There is a leader vehicle that controls speed and direction, and all following vehicles (whose braking and acceleration are precisely coordinated) respond to the leader vehicle's movement.



History

1972–73, the European ARAMIS project platooned 25 small transport vehicles travelling 30 centimetres apart on a French test track. The vehicles used ultrasonic and optical-range sensors.



The Prometheus Project in Europe (1980–1995) is another milestone in which car and truck manufacturers, technology companies, universities, and government joined forces to create intelligent vehicles that would operate in an advanced road system. The arrangement would include key areas such as communications, vehicle control, and artificial intelligence.

One of the companies significantly involved in this project was Volkswagen, which conducted highway-speed test runs. The vehicles had fully automated steering and longitudinal control. Although the project proved promising, it was eventually discontinued for political reasons.

Experiments continued in the United States to develop better platooning technologies. At UC Berkeley, the Partners for Advanced Transportation Technology (PATH) project began in 1986 as a collaboration between state and local governments and pioneered several intelligent systems.

The project remains operational today and currently operates platoons of three trucks at 14-foot (4.26-metre) gaps.

The Safe Road Trains for the Environment (SARTRE) project (2010–2012) was carried out in partnership with the European Commission, Ricardo UK Ltd, and Volvo. The project focused on designing intelligent systems for truck platooning that would also provide important environmental and safety benefits.

Peloton Technologies, a Silicon Valley company, in collaboration with Volvo and Peterbilt, developed a truck-platooning system that integrates safety, efficiency, and analytics for collision mitigation and adaptive cruise-control systems.

The system combines vehicle-to-vehicle (V2V) communications, radar-based active braking systems, and proprietary vehicle-control algorithms to electronically couple trucks.

Peloton conducted successful demonstrations in Detroit and looks forward to demonstration showcases in other regions.

In general, there are three types of vehicles: platoon-capable vehicles, non-platoon vehicles, and platoon vehicles.

A platoon-capable vehicle has all required hardware and software and can be either a non-platoon or a platoon vehicle.

A non-platoon vehicle, on the other hand, is not part of any platoon and travels under manual control, whereas a platoon vehicle is a member of a platoon.

A lane is reserved for platooning to distinguish non-platoon from platoon-capable vehicles. Only platoon-capable vehicles are permitted to join a vehicle platoon.

Modern communications, such as Bluetooth and wireless networks, GPS, radar-detection systems, as well as drive-by-wire steering and accelerator systems, allow computers to take control of cars. 5G communications can help with the volume of data that must be processed to make platooning a safe option.

Platoon communication must meet safety and reliability requirements to keep the platoon stable and support efficient platoon maneuver operations.

Supported maneuver operations include joining, leaving, merging, and splitting.

Joining and leaving refer to entering and exiting a platoon, respectively.

A merging operation means combining two platoons travelling in the same lane.

Platoon splitting is defined as separating one platoon into two smaller platoons.

Platoon members communicate with one another through periodic platoon data packets, maneuver request/response packets, and member-view packets.

Constant-spacing platoon;
Greater message exchange.

Variable spacing;
Worse platoon performance.

Platoon stability is achieved through the periodic exchange of platoon data packets. The platoon data packet is initiated by the platoon leader.

The packet contains the platoon identifier, lane identifier, sequence number, acceleration, speed, position, and sender address of the packet transmitter.

The receiving vehicle adjusts its position and then updates the sender address, speed, and acceleration fields in the platoon data packet and forwards it to the following vehicle.

The purpose of this speed and distance adjustment is to maintain a safe space gap to the vehicle ahead.

The platoon leader coordinates all platoon maneuvers. Maneuvers may occur at any time, and only one maneuver is allowed at a time.

Platoon followers must inform the platoon leader before carrying out any maneuver action.

First, a maneuver-request packet is sent from the initiating vehicle to the destination vehicle, possibly in multiple hops.

The initiating vehicle is the platoon member through which a new vehicle must join in a joining maneuver; the platoon member that must leave in a leaving maneuver; the platoon leader that intends to merge with another platoon in a merging maneuver; and the platoon leader in a splitting maneuver.

The destination vehicle is the platoon leader. The maneuver-request packet contains the maneuver identifier and the addresses of the initiating and destination vehicles.

After receiving the packet, a message is sent back to the initiating vehicle with information about the suitability of the platoon maneuver.

After completion of any maneuver and periodically thereafter, members are updated with the platoon member view.

The ordered sequence of vehicle identifiers in the vehicle platoon.

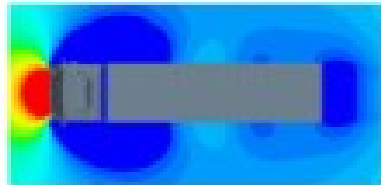
Potential advantages of vehicle platooning

It improves aerodynamic efficiency and performance, increases road capacity, and provides a more stable traffic flow.

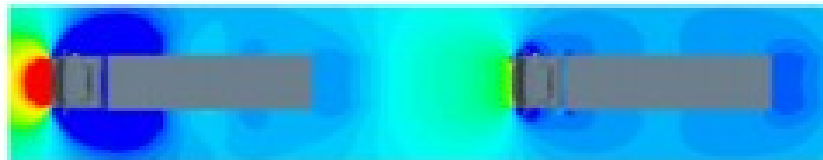
Because vehicles are drafting, it reduces fuel consumption.

Significant energy savings through aerodynamic drafting

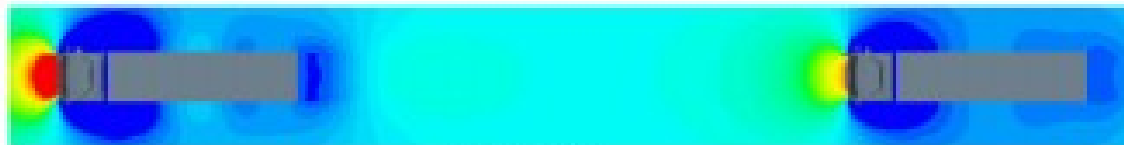
A more stable vehicle-following dynamic, reducing traffic-flow disturbances and providing additional energy and emissions savings



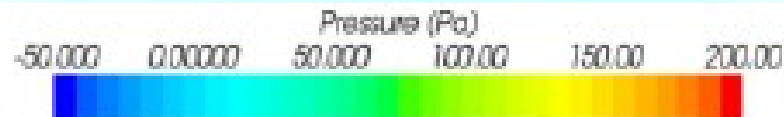
(i) No-traffic



(ii) 30 ft



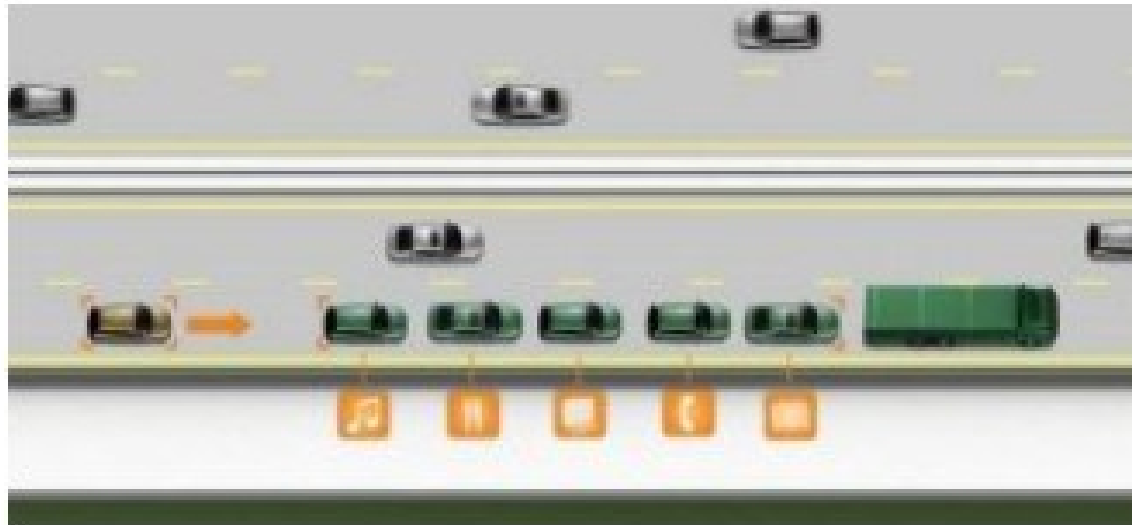
(iii) 90 ft



Potential safety improvement and accident reduction



Initially, each platoon vehicle would have a driver at all times, but eventually driverless (autonomous) vehicles will be able to automatically join platoons that can operate as long vehicle trains on a highway, and drivers would be free to enjoy the benefits of a driverless vehicle, such as being able to watch a movie or work.



Possible disadvantages of vehicle platooning

Not all highways can accommodate platoons.

Long platoons may prevent other vehicles from changing lanes when necessary.

A platoon will require a cultural change: giving up control of a motor vehicle, granting full control, and travelling very close together.

The system must be able to operate in hybrid mode, disabling platooning where it is not permitted.

Higher vehicle costs;

Who will be held liable in an accident involving a vehicle platoon.

If the vehicle ahead does not have time to react to an obstruction—for example, a vehicle turning in front of it leaves no room to stop—all other vehicles will likely be involved in the collision due to their close following distance.

Given the following distances, if the vehicle ahead drives over an object, all following vehicles will also drive over that object.

Autonomous vehicles greatly reduce the likelihood of this occurring.

Artificial intelligence to minimize occurrences.

Drivers must be able to leave the platoon at any time and join other platoons or take full control. This is because not every car will have the same final destination.

Safety failures may occur during this process.

Attacks on platoons

No computer system is 100% secure, and platoons can be hacked.

An attacker controlling a vehicle attempts to disrupt the platoon.

Outsider: has NO prior knowledge of the control law and only changes its movement.

Insider: modifies the control law and its movement.

Platoon Data Packet Injection: Data falsification and replay attacks are examples of this class. Data falsification is defined as changing the content of the platoon data packet and retransmitting it as if the message came from platoon members.

For example, a malicious agent may change the acceleration field in the platoon data packet from deceleration to acceleration. This can destroy platoon stability, possibly resulting in a collision.

In a replay attack, the malicious agent listens to a packet transmitted over the platoon communication medium, stores it, and retransmits it later as if it were a new packet.

Although the content of the platoon data packet is not modified in a replay attack, outdated information can mislead platoon members, possibly ruining platoon stability.

The synchronized Global Positioning System (GPS) clock or packet sequence numbers are used to prevent data-packet replay attacks, while cryptographic keys, digital certificates, and signatures eliminate data-packet falsification attacks.

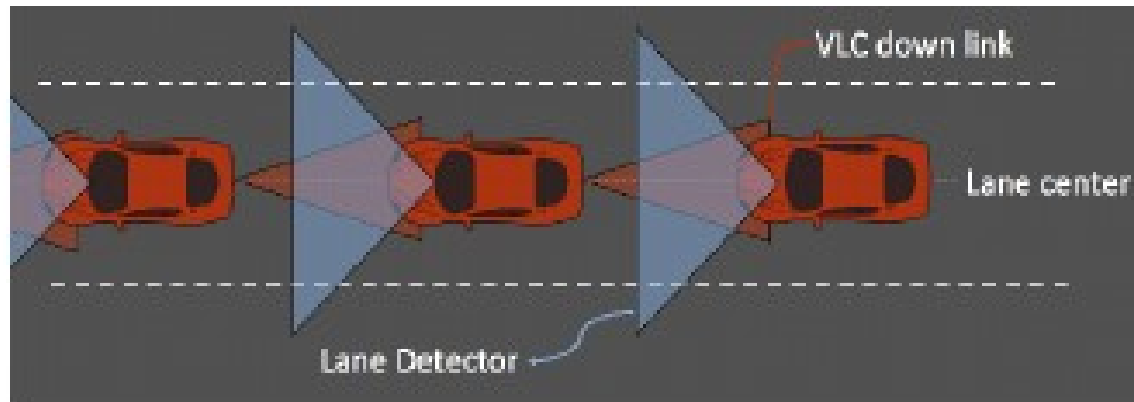
Platoon Channel Overhearing: a malicious actor listens to the platoon communication medium, collects private vehicle and platoon-operation information, and impersonates platoon members' identities. The information the malicious agent can obtain includes the route and final destination of autonomous vehicles, putting privacy at risk. For example, vehicle-rental companies or insurers may want to illegitimately follow vehicles to track passengers.

Platoon Jamming: The malicious actor blocks the platoon communication medium. Jamming occurs when the adversary receives information related to the vehicle platoon. The malicious actor aims to violate the medium access control (MAC) protocol and causes packet-delivery delays, endangering stable platoon operation.

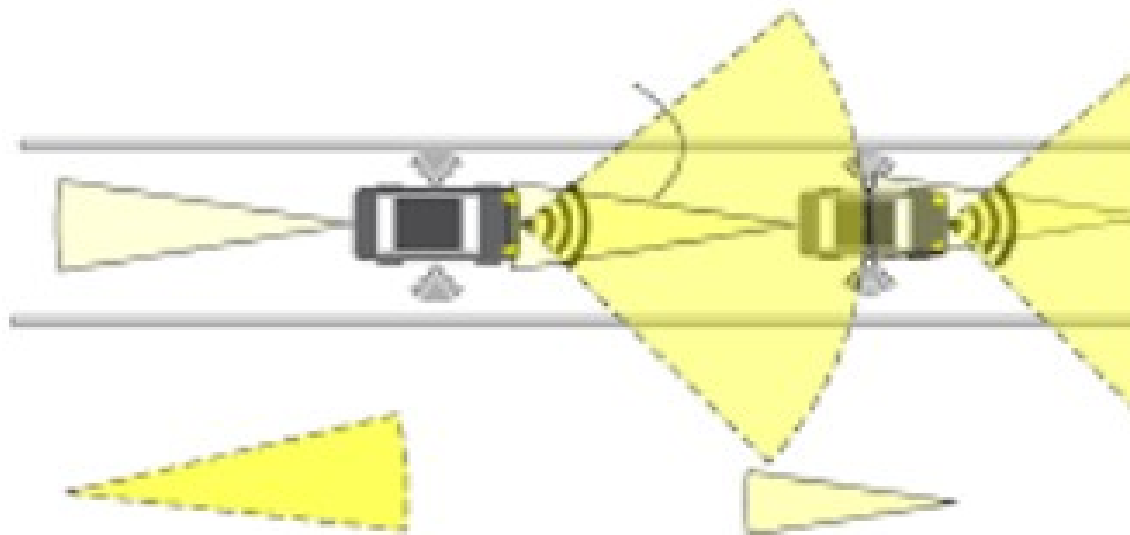
Platoon maneuver attack: the malicious actor generates a fake maneuver-request packet or a fake maneuver-response packet.

VLC platoon

VLC transmitters and receivers are placed at the front and rear of the vehicle.
VLC transmitters are connected to the vehicle's headlights and taillights.



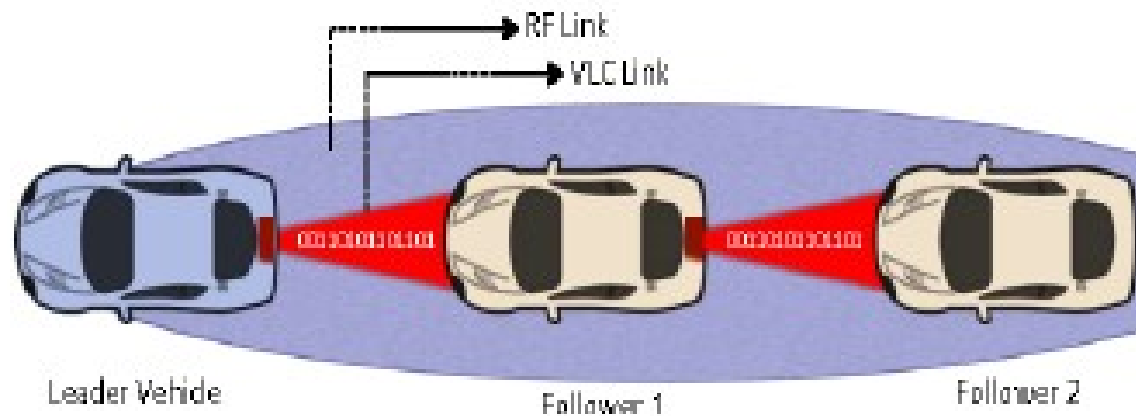
The transmission characteristics of taillights and headlights are different, resulting in an asymmetric communication link between successive vehicles in the platoon.



Sending messages from the platoon leader to all members via VLC is not possible because of its directionality and obstacles caused by other vehicles.

Weather conditions;

Hybrid RF/VLC use



Paving the way to Connected Automation - Cooperative Adaptive Cruise Control

<https://youtu.be/2-WoV8nKQUE>

Corda Wireless Exiting and Joining Platoon

<https://youtu.be/oeLh03gB8Gg>

Cohda Wireless S Curve Platooning @ The Bend Motorsport Park

<https://youtu.be/37dw6jsvsq8>

All aboard truck platooning

<https://youtu.be/VDTtatx6xLY>

Innovation Day in Hamburg - Use Case AUTOMATED

<https://youtu.be/wSmTgBQhMr8>

Truck Platooning: An Early Application of Cooperative Vehicle Automation

<https://youtu.be/qycc1nZQtLc>